MTH636 – Game Theory

Lecture 3: Dominated Strategies & Iterated Elimination

1 Strictly Dominated Strategies

Definition: Strictly Dominated Strategy

A strategy s_i of player i is **strictly dominated** by another strategy t_i of the same player if, for all possible strategy profiles of the other players $s_{-i} \in S_{-i}$, playing t_i always yields a strictly higher payoff than playing s_i. Mathematically:

$$u_i(s_i, s_{-i}) < u_i(t_i, s_{-i}) \quad \forall s_{-i} \in S_{-i}$$

Key Insight: Rationality Assumptions

- **Assumption 1:** A rational player will *never* play a strictly dominated strategy.
- **Assumption 2:** All players in a game are rational.

Let $S_{-i} = \prod_{j \neq i} S_j$ denote the set of all possible combinations of strategies for all players other than i.

1.1 Iterated Elimination of Strictly Dominated Strategies (IESDS)

Example: IESDS Process

Consider the following game:

$P_1 \setminus P_2$	L	M	R
T	(1, 0)	(1, 2)	(0, 1)
B	(0, 3)	(0, 1)	(2, 0)

Step 1: Look at Player 2's strategies (L, M, R).

- If P1 plays T, P2 gets 0 (L), 2 (M), 1 (R).
- If P1 plays B, P2 gets 3 (L), 1 (M), 0 (R).

Notice that for Player 2, strategy **M strictly dominates R** because $2 > 1$ and $1 > 0$. Since Player 2 is rational, they will never play R. We eliminate column R:

$P_1 \setminus P_2$	L	M
T	(1, 0)	(1, 2)
B	(0, 3)	(0, 1)

Step 2: Now look at Player 1's strategies (T, B) in the reduced game.

- T yields payoffs (1, 1).
- B yields payoffs (0, 0).

For Player 1, strategy **T strictly dominates B** ($1 > 0$ and $1 > 0$). Player 1 knows Player 2 is rational and won't play R. Thus, we eliminate row B:

$P_1 \setminus P_2$	L	M
T	(1, 0)	(1, 2)

Step 3: Back to Player 2. In this further reduced game:

- L yields 0.
- M yields 2.

Strategy **M strictly dominates L** ($2 > 0$). Eliminate column L.

$P_1 \setminus P_2$	M
T	(1, 2)

Result: The unique prediction is that Player 1 plays T and Player 2 plays M. This relies on Player 1 knowing that Player 2 is rational.

2 Common Knowledge and Complete Information

Definition: Common Knowledge

A fact is **Common Knowledge** among the players in a game if for any finite chain of players $i_1, i_2, \ldots, i_k$, the following holds: Player i_1 knows that Player i_2 knows that $\ldots$ Player i_k knows the fact.

Key Insight: Assumption 3

Rationality of the players is common knowledge.

Definition: Game of Complete Information

A game is called a game of **complete information** if the structure of the game (the players, their strategies, and their payoff functions) is common knowledge among all players.

Theorem: Order Independence of IESDS

The order of elimination of *strictly* dominated strategies does not matter, in the sense that the final game obtained after removing all strictly dominated strategies does not depend on the order of elimination.

3 Weakly Dominated Strategies

Definition: Weakly Dominated Strategy

A strategy s_i of player i is **weakly dominated** by another strategy t_i of the same player if:

1. $u_i(s_i, s_{-i}) \leq u_i(t_i, s_{-i}) \quad \forall s_{-i} \in S_{-i}$
2. $\exists t_{-i} \in S_{-i}$ such that $u_i(s_i, t_{-i}) < u_i(t_i, t_{-i})$

(i.e., t_i is always at least as good as s_i, and is strictly better against at least one strategy profile of the opponents).

3.1 Iterated Elimination of Weakly Dominated Strategies

Unlike strict domination, the order of elimination *does* matter for weakly dominated strategies.

Example: Order Dependence Example

Consider the following game:

$P_1 \setminus P_2$	L	C	R
T	(1, 2)	(2, 3)	(0, 3)
M	(2, 2)	(2, 1)	(3, 2)
B	(2, 1)	(0, 0)	(1, 0)

Let's observe different valid elimination sequences:

Sequence 1: Eliminate T, R, B, C

1. For P1, **M weakly dominates T** (since $2 > 1$, $2 = 2$, and $3 > 0$). Eliminate T.
2. For P2, against {M, B}, **L weakly dominates R** (payoffs 2,1 vs 2,0). Eliminate R.
3. For P1, against {L, C}, **M strictly dominates B** (payoffs 2,2 vs 2,0). Eliminate B.
4. For P2, against {M}, **L strictly dominates C** (payoffs 2 vs 1). Eliminate C.
5. **Result:** (M, L).

Sequence 2: Eliminate B, L, C, T

1. For P1, **M weakly dominates B** (since $2 = 2$, $2 > 0$, $3 > 1$). Eliminate B.
2. For P2, against {T, M}, **R weakly dominates L** (payoffs 3,2 vs 2,2). Eliminate L.
3. For P2, against {T, M}, **R weakly dominates C** (payoffs 3,2 vs 3,1). Eliminate C.
4. For P1, against {R}, **M strictly dominates T** (payoffs 3 vs 0). Eliminate T.
5. **Result:** (M, R).

Key Takeaway

While the Iterated Elimination of Strictly Dominated Strategies (IESDS) yields a unique result regardless of the order of elimination, the **Iterated Elimination of Weakly Dominated Strategies is order-dependent**. Different elimination paths can lead to entirely different predictions.